Exercise 1

1. SELECT Statement

```
SELECT *
FROM employees;
```

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	54000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laurel	Hall	IT	50000	2020-08-10	San Francisco

2. SELECT DISTINCT statement

```
SELECT DISTINCT departments
FROM employees;
```

department
IT
HR
Finance
Marketing

3. ORDER BY statement

```
SELECT      first_name,
            last_name
FROM        employees
ORDER BY    salary DESC;
```

first_name	last_name
Emily	Davis
Mike	Johnson
Robert	Wilson
John	Doe
Sarah	Brown
Daniel	Clark
David	White
Jessica	Moore
Laura	Hall
Jane	Smith

4. LIMIT statement

```
SELECT      first_name,
            last_name,
            salary
FROM        employees
ORDER BY    salary DESC
LIMIT       5;
```

first_name	last_name	salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000

5. WHERE statement

```
SELECT      first_name,
            last_name,
            department
FROM        employees
WHERE       department = 'IT' ;
```

first_name	last_name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6. AND statement

```
SELECT      *
FROM        employees
WHERE       department = 'Finance' AND salary > 58000 ;
```

id	first_name	last_name	department	salary	hire_date	city
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59000	2019-10-01	Houston

7. OR statement

```
SELECT      *
FROM        employees
WHERE       department = 'HR' OR 'Marketing' ;
```

id	first_name	last_name	department	salary	hire_date	city
2	Jane	Smith	HR	48000	2019-07-20	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

8. NOT statement

```
SELECT      first_name,
            last_name,
            department
FROM        employees
WHERE NOT   department = 'IT';
```

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. IN statement

```
SELECT      first_name,
            last_name,
            department
~~WHER~~
FROM        employees
WHERE       department IN ('HR', 'IT', 'Finance');
```

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

10. Combining conditions

```
SELECT    first_name,
          last_name,
          department,
          salary,
          city
FROM      employees
WHERE     department = 'IT',
AND       salary > 50000,
AND       City = 'New York';
```

first_name	last_name	department	salary	city
John	DOE	IT	55000	New York
Sarah	Brown	IT	53000	New York

11. Combining WHERE, AND, and ORDER BY

```
SELECT    first_name,
          last_name
FROM      employees
WHERE     department IN ('Finance', 'Marketing')
AND       salary > 52000
ORDER BY  salary DESC;
```

first_name	last_name
Mike	Johnson
Robert	Wilson
Daniel	Clark

12. Combining SELECT DISTINCT, WHERE and IN

```
SELECT DISTINCT   city
FROM              employees
WHERE             departments NOT IN ('IT', 'HR');
```

city
Los Angeles
San Francisco
Houston
Chicago

13. <u>Combining WHERE, NOT, AND, and ORDER BY</u>

```
SELECT      first_name,
            last_name,
            department,
            salary,
            hire_date
FROM        employees
WHERE NOT   department = 'Finance'
      AND   salary > 50000
ORDER BY    hire_date ASC;
```

first_name	last_name	department	salary	hire_date
Emily	Davis	IT	62000	2015-02-14
David	White	Marketing	52000	2016-04-10
Jessica	Moore	HR	51000	2018-05-22
John	Doe	IT	55000	2018-06-15
Sarah	Brown	IT	53000	2021-03-25
Daniel	Clark	Marketing	53000	2022-06-01

14. Combining WHERE, OR, IN and LIMIT

```
SELECT      first_name,
            last_name,
            department,
            city
FROM        employees
WHERE       city IN ('Chicago', 'Los Angeles')
      AND   department IN ('IT', 'Marketing')
LIMIT       3;
```

first_name	last_name	department	city
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago